Interpretable by Construction, Identifiable by Construction: Privacy Leakage of Shapelet-Based Anti-Money-Laundering Models

Anonymous Author(s)

ABSTRACT

Investigators and supervisors ask that anti-money-laundering (AML) models be explainable, and shapelet classifiers are a natural answer: they justify a decision by pointing to a short, human-readable pattern in an account's activity. We show that this explanation is frequently *a customer's own transaction history*. On SAML-D (9.5M transactions, 28 typologies) we train shapelet detectors on activity-matched weekly account series and hand the fitted model to an adversary holding ~30k candidate accounts. For the classic shapelet transform, 15% of released shapelets uniquely re-identify their source customer from shape alone, and 66% of those sources are flagged launderers, so the model also discloses who is under suspicion. For the state-of-the-art Random Dilated Shapelet Transform (RDST), which stores per-shapelet normalisation statistics, 86% of shapelets single out exactly one customer (≈827 customers) and reveal their exact weekly amounts and counts. The leak survives a realistic adversary whose amounts are off by ±5% (227 customers re-identified at 96% precision per 300 shapelets) and degrades only when the adversary misses a large share of transactions. It replicates on AMLworld (64% of RDST shapelets single out one customer). Standard membership-inference audits miss it (AUC ≈ 0.50–0.52); conventional models leak *whether* a customer was in training, if anything, while shapelet models leak *who* they are and *what* they did. Noise and k-anonymous averaging remove exact fingerprints at no accuracy cost but leave residual linkage, whereas *learned* shapelets copy no customer and show no memorisation even when overtrained; on SAML-D they also match RDST accuracy (PR-AUC 0.957 vs. 0.948), while on the harder AMLworld they cost accuracy (0.42 vs. 0.57). Interpretable financial models need a privacy review of their explanations, not only of their predictions.

1 INTRODUCTION

Transaction monitoring for AML is high-stakes, regulated, and increasingly machine-learned; deployed systems pair gradient-boosted trees with post-hoc explanations so that investigators can act on alerts [7]. Shapelets [1, 2] are an attractive alternative because the explanation *is* the model: a short subsequence whose presence drives the decision. Laundering typologies such as structuring, fan-in/fan-out and rapid pass-through are literally shapes in time.

The catch is that classic shapelets are subsequences copied from training series, and state-of-the-art variants such as RDST [4] additionally store the mean and standard deviation used to normalise each shapelet. Federated shapelet work notes that shapelets expose raw data [8], and privacy-aware matrix profiles defend shared time-series features against pattern inference [10]; explanation-privacy research shows that feature attributions leak membership [13]. To our knowledge, no prior work measures what a *released shapelet model* reveals about specific customers.

Contributions. (i) A threat model and attack suite—re-identification, amount reconstruction, membership and suspicion inference—for released shapelet models. (ii) An empirical study on two AML benchmarks showing large but *concentrated* leakage that aggregate membership inference misses, including disclosure of flagged customers (a tipping-off hazard), and mapping when it survives an imperfect adversary. (iii) A privacy–utility comparison of mitigations showing that learned shapelets attain near state-of-the-art shapelet accuracy without copying customer data.

2 RELATED WORK

Shapelets. Shapelets [1] classify time series by the presence of short discriminative subsequences; the shapelet transform [2] turns distances to the best k subsequences into features, learned shapelets [3] optimise them by gradient descent, and RDST [4] samples dilated, optionally normalised subsequences at scale and is among the strongest shapelet methods. Their appeal in finance is interpretability: shapelets have been proposed for interpretable early classification and financial forecasting. **ML for AML.** Production AML monitoring relies on gradient-boosted trees over aggregates with post-hoc explanations [7]; research benchmarks include SAML-D [6] and AMLworld [9], with graph and sequence models as strong learners. Supervisors increasingly expect explainable alerts, which motivates model-intrinsic explanations such as shapelets. **Privacy of models and explanations.** Membership inference [11] is the standard audit of training-data leakage; strong audits emphasise low false-positive rates [12]. Explanations themselves leak membership [13]. For time series, federated shapelet learning protects raw subsequences with secure computation [8], and privacy-aware matrix profiles defend against sensitive-pattern inference [10]; neither quantifies what a released shapelet model reveals about individuals, which is our focus.

3 THREAT MODEL

A bank trains a shapelet AML detector and shares the fitted model with a vendor, auditor, regulator sandbox, or consortium partner, or loses it to theft. The model contains its shapelets (and, for RDST, their normalisation statistics). The adversary also holds candidate account series, e.g. a partner institution's data or a leaked extract, containing training members and non-members. We study both an idealised adversary (exact copy, full account list) and a realistic one whose amounts are perturbed, who sees only part of each account's transactions, and whose list holds only some accounts. Goals: (a) *re-identify* the customer behind a shapelet; (b) *reconstruct* that customer's amounts and counts; (c) infer *membership* and, from the shapelet's class, *suspicion status*.

Table 1: Detection on held-out accounts and leakage of the released model (mean ± std over 3 seeds). Re-ID: share of released shapelets that match *only* their source customer among 29,946 candidate accounts.

Model	PR-AUC	R@1%FPR	MIA AUC	Re-ID (%)
CatBoost (aggregates)	0.983 ± 0.001	0.842 ± 0.031	0.504 ± 0.002	n/a
MiniRocket	0.955 ± 0.002	0.637 ± 0.044	0.508 ± 0.003	n/a
Shapelet transform	0.878 ± 0.007	0.493 ± 0.033	0.500 ± 0.002	15.0 ± 2.2
RDST	0.948 ± 0.006	0.631 ± 0.021	0.520 ± 0.001	85.6 ± 1.6
RDST, norm.-only, no stats	0.925 ± 0.005	0.515 ± 0.015	0.512 ± 0.000	74.9 ± 1.9
Learned shapelets	0.957 ± 0.004	0.667 ± 0.017	0.500 ± 0.000	0.0

4 EXPERIMENTAL SETUP

Data. SAML-D [6]: 9.5M synthetic transactions over 46 weeks, 0.10% suspicious, 17 suspicious and 11 normal typologies. Each account becomes a 6-channel weekly series (log outgoing/incoming amount, outgoing/incoming count, cross-border amount, cash amount). Accounts involved in any laundering transaction are positive (7,902). Normal accounts are sampled 3:1 *matched on number of active weeks*; unmatched normals are mostly single-week accounts that make detection trivial. The 29,946 accounts are split 50/50 by account into members (train) and non-members (test), stratified by label; results average 3 seeds. **AMLworld** HI-Small [9] (IBM's NeurIPS 2023 benchmark) spans only ~10 days, so we use 6-hour bins (40 steps); 23,409 activity-matched accounts, 5,862 positive.

Detectors. CatBoost on 40 aggregate features plus the flattened series (the industry baseline); MiniRocket [5] with logistic regression; the binary shapelet transform (5,000 random non-constant z-normalised candidates, quality = class separation of the min-distance profile, top-200 with self-similarity removal) with ExtraTrees; RDST (1,000 shapelets) with ExtraTrees; and learned shapelets [3] (144 shapelets, soft-min distance, random initialisation).

Attacks. *Re-ID*: min (dilated) sliding distance from each released shapelet to every candidate account; we report the share of shapelets whose source is the *only* account within ε (unique match) and top-1 accuracy. *Reconstruction*: for RDST, raw window = values × std + mean. *MIA*: score = −min distance to any shapelet; for baselines, the standard confidence attack. *Realistic adversary*: we rebuild the adversary's series from the raw transactions with log-normal amount noise (σ=0.05) and each transaction kept with probability 0.8 or 0.5, subsample the candidate list, and claim an identity only if the nearest account is ≥ 2× closer than the runner-up.

5 RESULTS

Leakage is severe but concentrated. Table 1: on SAML-D only the shapelet models show leakage; CatBoost and MiniRocket, which store no training subsequences, give no membership signal. Because just a few hundred of ~15k members contribute shapelets, aggregate MIA AUC stays near 0.5, yet those contributors are uniquely identifiable: an audit based on standard MIA would declare these models safe.

Shape alone identifies. Z-normalised shapelet-transform shapelets carry no amounts, yet 15% match only their source among ~30k accounts (top-1: 19%).

Table 2: Mitigations (shapelet transform; mean over seeds). Unique: shapelet matches only its source; Top-1: source is the nearest account.

Release	PR-AUC	Unique (%)	Top-1 (%)
As trained	0.878 ± 0.007	15.0 ± 2.2	19.3 ± 1.6
Noise σ=0.1	0.879 ± 0.007	0.0 ± 0.0	10.8 ± 2.5
Noise σ=0.25	0.877 ± 0.007	0.0 ± 0.0	3.2 ± 0.5
Noise σ=0.5	0.876 ± 0.005	0.0 ± 0.0	0.2 ± 0.2
Noise σ=1.0	0.870 ± 0.004	0.0 ± 0.0	0.0 ± 0.0
k-anon. mean, k=5	0.877 ± 0.007	0.3 ± 0.2	13.5 ± 0.4
k-anon. mean, k=20	0.876 ± 0.005	0.2 ± 0.2	12.5 ± 2.1
k-anon. mean, k=50	0.876 ± 0.006	0.0 ± 0.0	10.7 ± 1.3

Table 3: Re-identification by a realistic adversary (SAML-D, 300 sampled RDST shapelets, mean over seeds). The adversary's copy of the data has perturbed amounts and/or misses transactions, and holds all or half of the accounts; it claims an identity only if the nearest account is at least 2× closer than the runner-up. Cust.: customers correctly re-identified; Prec.: share of claims that are correct.

	All accounts		Half of accounts	
Adversary's view	Cust.	Prec.	Cust.	Prec.
Exact copy	251	94%	125	86%
Amounts ±5%	227	96%	115	86%
80% of txns, ±5%	71	78%	36	62%
50% of txns, ±5%	10	27%	5	17%

Table 4: Second dataset: AMLworld HI-Small (23,409 activity-matched accounts, 6-hour bins; mean over 3 seeds). MIA: standard membership-inference AUC. Re-ID: share of shapelets matching only their source; Cust.: customers singled out per model.

Model	PR-AUC	MIA AUC	Re-ID (%)	Cust.
CatBoost	0.619 ± 0.004	0.625 ± 0.004	n/a	n/a
MiniRocket	0.475 ± 0.003	0.586 ± 0.003	n/a	n/a
Shapelet transform	0.408 ± 0.016	0.499 ± 0.002	6.0 ± 1.1	–
RDST	0.571 ± 0.001	0.507 ± 0.002	64.5 ± 1.2	621 ± 10
RDST, norm.-only, no stats	0.546 ± 0.005	0.498 ± 0.002	40.4 ± 1.7	391 ± 14
Learned shapelets	0.389	0.500	0.0	0

State of the art leaks more. RDST is the most accurate shapelet model here but stores normalisation statistics, so 86% of its shapelets reconstruct an exact, unique slice of one customer's history: about 827 customers per model, each with a median of 13 exact non-zero weekly amount/count cells (median peak weekly amount ≈ £25k). Withholding the statistics and forcing every shapelet to be normalised barely helps: 75% of shapelets still match a single customer (PR-AUC drops to 0.925), because a multi-channel dilated shape is itself a near-unique fingerprint—and once an account is re-identified, the adversary's own data supplies its amounts.

Table 5: Pushing learned shapelets to memorise (SAML-D, seed 0). Nearest-is-member: share of shapelets whose nearest account is a training member, vs. the members' share of the pool (chance). Claims: confident re-identifications ($d_2 \geq 2d_1$).

Training	PR-AUC	Nearest-is-member (chance)	Claims
30 epochs, full train	0.957	50% (50%)	0
200 epochs, full train	0.952	50% (50%)	0
200 epochs, 1k train	0.928	4% (3%)	0
400 epochs, 300 train	0.895	0% (1%)	0

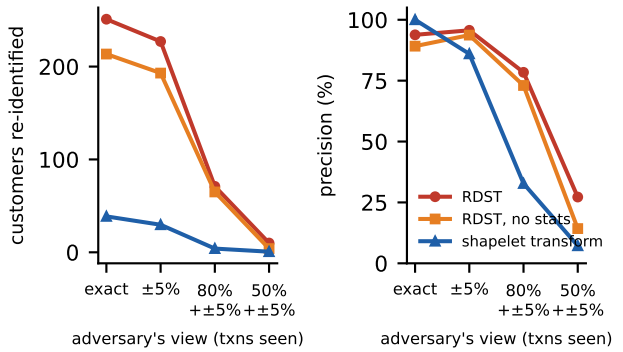


Figure 1: Realistic adversary on SAML-D (full account list; mean over seeds). Amount noise barely matters; missing transactions are what stops the attack.

A realistic adversary still succeeds (Table 3, Figure 1). Perturbing every amount by ±5% barely matters: the adversary re-identifies 227 customers (62 flagged) at 96% precision per 300 RDST shapelets; with stored statistics withheld, 193 at 94%. With half the account list, it finds about half as many, still mostly correctly. The limit is coverage of each account's activity: seeing 80% of transactions leaves 71 correct identifications at 78% precision, while at 50% precision falls to 27%. The threat is therefore real for parties that see an account's full activity (vendors and auditors given bank data, insiders, breaches) and much weaker for outsiders with a partial view.

It replicates on a second benchmark (Table 4). On AML-world, with 6-hour bins and a much harder task, RDST is the best time-series detector (PR-AUC 0.57; CatBoost 0.62) and 64% of its shapelets single out one customer (≈621 per model, 164 flagged); shape-only shapelets leak less (6%). Learned shapelets again copy no one, but here they trail RDST (PR-AUC 0.39, or 0.42 with shorter patterns and longer training, where 13% of shapelets coincide with generic windows shared by nearly every account but none singles out a customer), so privacy is not free on every dataset. Here CatBoost and MiniRocket overfit and do leak membership (MIA AUC 0.62 and 0.59), which sharpens the contrast: a conventional model leaks *whether* a customer was used, a shapelet model leaks *who* they are and *what* they did, and the standard audit sees only the former.

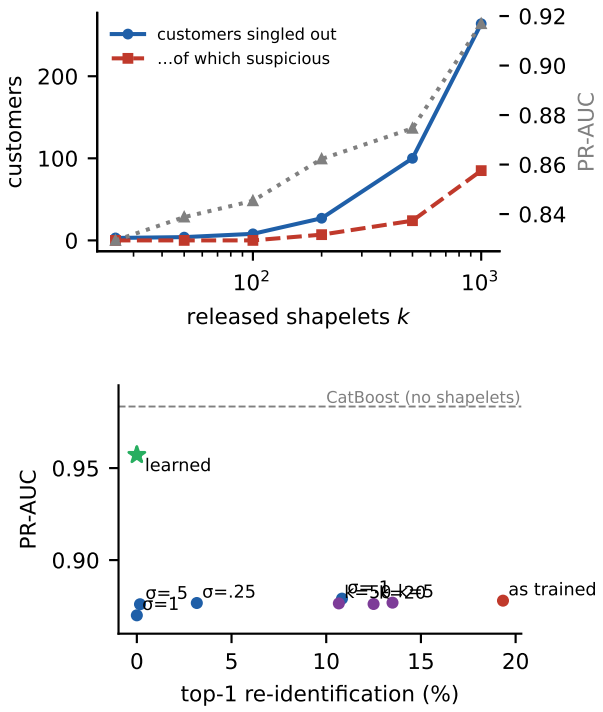


Figure 2: Top: customers uniquely exposed (and how many are flagged launderers) grow with the number of released shapelets k, while accuracy saturates (shapelet transform, seed 0). Bottom: privacy–utility of releases (top-1 re-identification vs. PR-AUC; seed mean).

The model discloses suspicion. 66% of shapelet-transform sources and 217 RDST-identified customers are positives. Releasing the model therefore reveals which customers were flagged, in tension with suspicious-activity-report confidentiality and anti-tipping-off rules.

Mitigations (Table 2). Gaussian noise on shapelet values removes exact fingerprints at no accuracy cost, but nearest-neighbour linkage persists until $\sigma \geq 0.5$. Replacing each shapelet by the mean of its k nearest training windows from distinct accounts (k-anonymous release) stops unique matching but keeps the source's neighbourhood linkable. Learned shapelets dominate: PR-AUC 0.957 (shapelet transform: 0.878; CatBoost: 0.983), with no learned shapelet within 10^{-2} of any real account (median nearest distance 0.38). Pushing them to memorise (Table 5)—up to 400 epochs on as few as 300 training accounts—does not change this: in every setting the share of learned shapelets whose nearest account is a training member equals the members' share of the pool (chance), and the adversary makes 0 confident claims.

6 DISCUSSION AND LIMITATIONS

SAML-D is synthetic and activity-matched sampling makes detection optimistic; however, the leakage mechanism (copied subsequences, stored statistics) does not depend on the data. Our shapelet transform is a faithful simplification (AUC-based quality instead

of information gain) because a standard implementation degenerated on our sparse series. Our realistic-adversary results bound the threat: it needs a near-complete view of each targeted account. A stronger, per-example membership audit (e.g. shadow-model likelihood-ratio attacks) may detect some of this leakage; our claim concerns the standard aggregate audit. Noise and learned shapelets carry no formal guarantee; differentially private shapelet selection and DP-SGD training are natural next steps.

Ethics. All experiments use synthetic data (SAML-D, AMLworld); no real customer is exposed. We describe the attack to motivate audits and defences that are cheap to adopt (learned or noised shapelets), and recommend that institutions treat released shapelets as personal data under their data-protection obligations.

Recommendations. Treat shapelets as personal data; do not rely on withholding normalisation statistics; prefer learned or noised shapelets, checking their accuracy cost per dataset; and audit interpretable financial models with *per-explanation* re-identification, not only aggregate membership inference.

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